

PORTLAND TROUTDALE, OR, USA

WMO: 726985

Lat: 45.551N

Lon: 122.409W

Elev: 9

StdP: 101.22

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 24242

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.7	-1.5	-11.8	1.4	-1.4	-8.4	1.8	0.4	14.6	3.0	13.8	3.3	7.9	90	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	33.1	19.6	31.1	19.2	28.9	18.4	20.9	30.5	20.0	29.2	19.0	27.5	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.5	22.9	16.6	11.8	22.5	15.9	11.3	22.1	60.1	30.8	56.8	29.1	53.7	27.5	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.3	10.1	-5.2	37.7	2.4	1.9	-6.9	39.1	-8.3	40.2	-9.6	41.3	-11.4	42.7		
(5)			-6.3	22.4	2.5	1.0	-8.1	23.0	-9.5	23.6	-10.9	24.2	-12.7	24.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	5.1	6.5	8.7	11.1	14.5	17.4	20.7	20.8	17.8	12.8	8.4	5.0	(6)	(7)
	DBStd	6.38	3.28	2.91	2.91	3.08	3.40	3.25	2.77	2.64	3.05	2.88	3.24	3.31	(7)	(8)
	HDD10.0	572	153	103	59	21	3	0	0	0	0	8	67	159	(8)	(9)
	HDD18.3	2384	409	333	297	218	127	54	8	7	46	172	299	414	(9)	(10)
	CDD10.0	1463	2	3	20	54	143	223	332	336	233	96	18	2	(10)	(11)
	CDD18.3	234	0	0	0	2	9	27	81	85	29	1	0	0	(11)	(12)
	CDH23.3	2654	0	0	0	22	150	357	888	869	357	11	0	0	(12)	(13)
	CDH26.7	1027	0	0	0	3	43	139	367	359	116	1	0	0	(13)	(14)
(14) Wind	WSAvg	3.1	5.6	4.6	3.4	2.6	2.0	1.9	1.7	1.6	1.9	2.6	4.3	5.2	(14)	(15)
(15) Precipitation	PrecAvg	1458	204	151	154	127	98	72	21	26	53	135	224	221	(15)	(16)
	PrecMax	1786	384	341	259	254	161	151	81	119	134	282	468	420	(16)	(17)
	PrecMin	1113	73	13	53	55	13	15	2	0	0	36	79	122	(17)	(18)
	PrecStd	175	71	68	48	44	39	30	17	23	31	63	90	69	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	15.5	21.2	26.3	30.7	34.6	36.9	36.0	32.7	24.5	18.2	15.1	(19)	(20)
		MCWB	11.7	10.8	13.6	15.7	18.0	20.2	21.1	20.3	18.6	15.3	14.5	12.7	(20)	(21)
	2%	DB	12.4	13.4	17.8	22.6	27.6	30.8	33.0	33.0	29.9	22.2	16.2	12.5	(21)	(22)
		MCWB	10.1	9.1	11.3	13.8	16.7	19.2	19.9	19.6	17.5	14.2	12.9	10.7	(22)	(23)
	5%	DB	11.1	12.2	15.7	19.3	24.2	27.4	30.8	30.9	27.5	19.7	14.3	11.0	(23)	(24)
		MCWB	9.0	8.6	10.2	12.0	15.4	17.6	19.5	19.3	17.3	13.5	11.7	9.2	(24)	(25)
	10%	DB	9.5	10.9	13.7	17.2	21.6	24.4	28.0	28.0	24.9	17.7	12.9	9.1	(25)	(26)
		MCWB	7.6	7.8	9.5	11.2	14.3	16.3	18.5	18.5	16.3	12.9	10.6	7.4	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	11.9	14.2	16.4	19.2	21.1	22.9	22.1	20.4	17.7	15.6	13.1	(27)	(28)
		MCDB	14.3	13.9	20.3	25.0	28.9	32.0	33.7	32.0	29.8	21.6	17.2	14.4	(28)	(29)
	2%	WB	10.5	10.3	12.2	14.3	17.4	19.6	21.0	20.8	19.0	15.8	13.4	10.9	(29)	(30)
		MCDB	12.1	12.4	16.0	21.2	25.7	29.4	30.3	30.4	27.0	19.3	15.3	12.4	(30)	(31)
	5%	WB	9.1	9.1	11.1	12.9	16.1	18.0	19.9	19.8	17.9	14.8	12.1	9.2	(31)	(32)
		MCDB	10.6	11.3	14.4	18.0	23.0	26.0	29.2	28.8	25.0	17.9	13.9	10.6	(32)	(33)
	10%	WB	7.9	8.0	10.1	11.7	14.9	16.9	18.9	18.9	17.0	13.9	11.1	7.8	(33)	(34)
		MCDB	9.2	10.1	13.3	15.8	20.5	23.5	27.4	27.0	23.1	16.9	12.9	9.0	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	6.3	8.2	9.5	10.5	11.0	13.2	13.4	12.4	8.7	5.6	4.4	(35)	(36)
		MCDBR	7.0	8.3	12.0	14.9	16.6	17.7	18.4	18.6	17.4	12.6	6.9	6.1	(36)	(37)
	5% WB	MCWBR	5.2	5.2	6.8	7.9	7.8	7.5	7.1	7.1	7.3	6.5	4.5	4.8	(37)	(38)
		MCWBR	6.7	7.1	10.0	12.6	14.4	15.9	16.6	16.5	15.2	9.6	6.1	5.9	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub	taub	0.320	0.316	0.329	0.365	0.378	0.361	0.351	0.358	0.345	0.335	0.319	0.325	(39)	(40)
		taud	2.525	2.549	2.526	2.389	2.375	2.438	2.487	2.488	2.502	2.539	2.546	2.474	(40)	(41)
	Edn at Noon	Edn at Noon	793	873	907	895	890	905	910	892	877	836	787	739	(41)	(42)
		Edh at Noon	66	77	90	112	117	111	104	101	91	77	64	63	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg	RadAvg	1.17	2.02	2.96	4.32	5.48	6.07	6.88	5.87	4.38	2.54	1.37	0.95	(43)	(44)
	RadStd	RadStd	0.19	0.28	0.38	0.39	0.51	0.54	0.46	0.39	0.37	0.31	0.18	0.14	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page